

Listening Between the Lines

Analyzing Gendered Conversational Dynamics in Indian Podcasts

Project Documentation — v2 Features and Dyad Analysis

Team: Bhavini Sai Mallu, Sameeksha Rao

Prepared: September 2026

About This Document

This document reports the second version (v2) of the project's linguistic feature analysis, in the order the scripts were built. Each section describes the purpose of a script, the methodology it implements, and the output it produced on the 418-episode podcast dataset. It is a companion to Project_Documentation.docx (the v1 documentation), which covers the first-version lexicon features, the balanced 200-episode sample, and the dialogue-act analysis; references to "v1 Section N" point to that document.

It covers: `src/compute_bias_features_v2.py` (Section 1, per-speaker hedge, booster and politeness features adapted to speech), the precision audits built with `src/make_audit_sample.py` (Section 2), `src/episode_dyads.py` (Section 3, gender dyads over all 418 episodes), `src/dyad_analysis_v2.py` (Section 4, episode-level dyad dataset), `src/stat_analysis/kruskal_wallis_test_v2.py` (Section 5), and `src/stat_analysis/Mann_Whitney_test_v2.py` and `src/stat_analysis/within_host_test_v2.py` (Section 6, pairwise dyad tests and the guest-gender test within host).

1. `src/compute_bias_features_v2.py`

1.1 Purpose

The manual audit of the first version (v1 Section 5) showed that the hand-built lexicons in `compute_bias_features.py` (v1 Section 1) were unreliable: only 12 of 50 sampled "directive" matches were real directives. This script replaces the four hand-built lexicons with two published resources and adapts them to spontaneous podcast speech using repeated precision audits (Section 2). It produces one row of conversational style features per speaker per episode, over all 418 collected episodes rather than the balanced 200-episode sample.

The two resources are Hyland's (2005) metadiscourse lexicon of hedges and boosters, and the ConvoKit PolitenessStrategies classifier (Danescu-Niculescu-Mizil et al., 2013), which marks 21 politeness strategies (gratitude, apologizing, deference, factuality, direct questions, 1st/2nd person, and others) using a dependency parse of each turn.

1.2 Methodology

- Input: the 418 WhisperX transcripts with diarized speakers, speaker gender, and HOST/GUEST labels (`data/outputs/whisperx/*_whisperx_diarized.gender.host_guest.json`).
- Turns: consecutive segments from the same speaker are merged into one turn before scoring. Segments without a diarized speaker are dropped, and speech in the first and last 60 seconds of each episode (intro/outro monologues and sign-off thank-yous) is excluded.
- Hyland hedges and boosters (`data/lexicons/hyland_2005_hedges_boosters.json`; 101 hedges, 64 boosters): all terms are compiled into one case-insensitive, whole-word regular expression, longest term first. Each stretch of text is matched at most once, so "no doubt" counts as a booster and does not also count "doubt" as a hedge.

- Spoken-speech adjustments to Hyland, stored in the same file under `spoken_adjustments`: (1) discourse-marker phrases such as "you know", "I don't know", "talk about" and "think about" are matched and discarded, so they cannot count as hedges or boosters; (2) "I think" is counted as a hedge, following Prokofieva & Hirschberg (2014) on hedging in conversation; (3) terms that audited as mostly literal in speech are dropped, e.g. hedges about, around, would, could, should, feel, quite and boosters know, think, thought, find, show, always, never, must, true, believe (with their inflected forms).
- ConvoKit: each episode's turns are loaded into a ConvoKit Corpus, parsed with spaCy (`en_core_web_sm`), and scored with `PolitenessStrategies` at the marker level, so every occurrence is counted, not just whether a strategy appears in a turn.
- ConvoKit adjustments (`data/lexicons/convokit_adjustments.json`): Factuality no longer counts "really" or "the point"; Deference no longer counts "good"; and `Direct_question` only counts a wh-question when its sentence ends with "?" and is the last sentence of the turn, so that the other speaker responds next. The reasons and audit counts for each change are recorded in the JSON files and summarized in Section 2.
- For each speaker in each episode, the script writes the word count, turn count, and for every feature the raw count and the rate per 1,000 words. ConvoKit features also get a turn rate: the share of the speaker's turns with at least one occurrence.
- Alongside the features, it writes a long table of which terms or markers produced each count (for tracing any feature back to its words), and a random sample of 50 hits per feature with the surrounding turn text, used for the precision audits.

1.3 How It Was Run

The script runs on the cluster as a CPU-only SLURM job (2 cores, 8 GB, about 21 minutes for 418 episodes) in the convokit conda environment:

```
sbatch sbatch/run_bias_features_v2.sbatch
```

which runs:

```
python src/compute_bias_features_v2.py --input_glob
"data/outputs/whisperx/*_whisperx_diarized.gender.host_guest.json" --out_csv
results/features/speaker_features_v2.csv --terms_csv
results/features/speaker_feature_terms_v2.csv --audit_csv
results/features/audit_sample_v2.csv --audit_n 50
```

Both sets of adjustments are on by default; `--no_spoken_adjustments` and `--no_convokit_adjustments` run the lexicons as published.

1.4 Output

Output file: `results/features/speaker_features_v2.csv` — 836 rows × 73 columns (one HOST row and one GUEST row for each of the 418 episodes), with no missing values.

File	Rows	Contents
------	------	----------

speaker_features_v2.csv	836	Features per (episode, speaker)
speaker_feature_terms_v2.csv	155,750	Count of each term/marker per (episode, speaker, feature)
audit_sample_v2.csv	1150	50 random hits per feature, with turn text

Columns produced:

- episode_id, speaker_id, role, gender, word_count, turn_count.
- hy_hedge_count, hy_hedge_per_1k, hy_booster_count, hy_booster_per_1k (Hyland).
- For each of the 21 ConvoKit strategies: ck_<strategy>_count, ck_<strategy>_per_1k, ck_<strategy>_turn_rate (e.g. ck_factuality_per_1k, ck_direct_question_turn_rate).

1.5 Dataset Composition of the Output

Role	Female	Male	Unknown	Total
HOST	125	293	0	418
GUEST	128	288	2	418

Guests speak far more than hosts: the median guest speaks 8,833 words per episode, against 2,388 for the median host. The two guests with unknown gender are resolved by manual overrides in Section 3.

1.6 Descriptive Statistics

Rates per 1,000 words for the five features carried into the dyad analysis (Sections 4-6), by role, with the words that account for most of each feature's hits:

Feature	Host mean / median	Guest mean / median	Top terms (share of hits)
Hedges	5.55 / 5.00	6.51 / 5.36	i think 41%, maybe 11%, sometimes 8%
Boosters	4.05 / 3.73	4.24 / 3.56	really 32%, actually 22%, of course 10%
Factuality	1.32 / 1.15	1.21 / 0.91	actually 71%, in fact 14%, honestly 7%
Deference	0.26 / 0.00	0.06 / 0.00	interesting 39%, nice 31%, great 22%
Direct questions	4.61 / 3.92	0.29 / 0.17	what 56%, how 26%, why 14%

The spoken-speech exclusions matter: 30,742 phrase matches were discarded ("you know" alone is 61% of them), which is about as many as all counted hedges. Direct questions are almost entirely a host behavior once rhetorical questions are removed. Deference is rare: most speakers never use it, so its median is 0.

1.7 Notes & Limitations

- Word lists cannot read context. The largest remaining source of error is "actually" used as filler ("thank you for actually being here"); it is 71% of Factuality hits and 22% of booster hits. It was kept because it is correct in roughly three quarters of audited uses.
- ConvoKit's strategies were designed on written requests (Wikipedia and Stack Exchange); podcast speech has more fillers, rhetorical questions, and backchannels, which is why the adjustments above were needed.
- Deference in speech is mostly acknowledgement ("Interesting.", "Nice.") rather than explicit praise, so it is better described as acknowledgement.
- Direct_question requires the question to end the turn, which removes about half of the real questions along with the rhetorical ones (Section 2). Rates are comparable across groups but undercount questions in absolute terms. Guest direct questions are not reliable and are not used.
- Diarization errors that merge two speakers into one turn can attribute one person's words to the other.

2. Precision Audits of the v2 Features (src/make_audit_sample.py)

2.1 Purpose

The v1 lexical audit (v1 Section 5) showed that a feature can look plausible and still be mostly wrong. Before using the v2 features, each one was checked by hand: a random sample of the words the script counted was read in context, and each hit was marked correct or not. The results drove the adjustments in Section 1.2, and the audit was repeated after each change.

2.2 Methodology

- compute_bias_features_v2.py samples 50 random hits per feature (reservoir sampling) into results/features/audit_sample_v2.csv.
- make_audit_sample.py selects the features being audited, rebuilds each hit's context from the full turn with the matched words in [[]], and adds a yes/no question for each feature. For example, for hedges: "Does [[]] express uncertainty, tentativeness, approximation, or reduced commitment?"; for direct questions: "Is this a direct wh-question actually asked of someone?"
- Each hit is labeled 1 (correct) or 0, with a short note for every 0. Precision is the share of hits labeled 1. Rhetorical and self-answered questions, reported speech, literal uses ("make sure", "the point where"), and filler uses count as 0.
- The labels were made by an AI assistant (Claude) following the per-feature questions, and are marked as such in the labeled_by column of each audit file. They have not yet been checked by a human annotator (see 11.6).

2.3 How It Was Run

```
python src/make_audit_sample.py --features  
hy_hedge,hy_booster,ck_factuality,ck_direct_question,ck_deference --out  
results/audit/audit_v2e_to_label.csv
```

2.4 Audit Rounds

Each round sampled fresh hits from the pipeline as it stood at the time (50 per feature).

Precision per round:

Round	Pipeline	Hedge	Boost.	Fact.	Def.	Q	Change made after
1 (v2)	Published lists + discourse-marker exclusions, "I think" as hedge	0.46	0.34	0.44	0.66	0.44	Drop Hyland terms that were mostly literal (about, would, feel, know, always, thought, ...); drop Factuality "really", Deference "good"; questions must end with "?"
2 (v2b)	After round 1 changes	1.00	0.70	0.84	0.72	0.32	Drop booster "believe" (mostly religious or others' beliefs)
3 (v2c)	After round 2 changes	0.84	0.74	0.76	0.84	0.58	Questions must end the turn; drop Factuality "the point" (0 correct in every round)
4 (v2d)	Final pipeline	—	—	—	—	0.86	None (direct questions only)
5 (v2e)	Final pipeline, fresh sample	0.86	0.68	0.68	0.80	0.86	None

Hedge = hedges, Boost. = boosters, Fact. = factuality, Def. = deference, Q = direct questions.

Round 1 also covered gratitude (0.88) and apologizing (0.80), which were accurate and needed no change; they are not used in the dyad analysis. Round 4's direct-question hits are the same 50 as round 5's.

2.5 Final Precision

For hedges, boosters and deference, rounds 3 and 5 sample the same final lexicon, and factuality only lost "the point", so they are pooled (about 100 hits each; round 3's "the point" hits removed). Direct questions use round 4, the only sample after its last change.

Feature	Hits	Precision (95% CI)	Host	Guest
---------	------	--------------------	------	-------

Hedges	99	0.86 (0.78-0.91)	0.76 (n=21)	0.88 (n=78)
Boosters	100	0.71 (0.61-0.79)	0.56 (n=27)	0.77 (n=73)
Factuality	95	0.76 (0.66-0.83)	0.65 (n=20)	0.79 (n=75)
Deference	98	0.82 (0.73-0.88)	0.92 (n=53)	0.69 (n=45)
Direct questions	50	0.86 (0.74-0.93)	0.97 (n=40)	0.40 (n=10)

All five features are between 0.71 and 0.86 overall. Two findings shaped the dyad analysis: guest direct questions stayed unreliable (mostly rhetorical questions inside stories), so direct questions are analyzed for hosts only; and booster and deference precision differs between hosts and guests, so the analysis compares each role separately across dyads rather than subtracting guest rates from host rates (Section 4.2). In the audit samples, precision for female and male speakers was similar, but there were only 5-16 female speakers per feature.

2.6 Notes & Limitations

- With 50 hits, a single round's precision is uncertain by about ± 0.10 ; changes between rounds smaller than that are within noise.
- The rule that questions must end the turn was chosen using the round 3 labels, which overstates how well it works on those labels (0.88). Round 4 checks it on new hits (0.86).
- Labels were made by an AI assistant and one pass only; a human check of a subset (for example 50 hits per feature, with inter-annotator agreement) would make these numbers citable.
- The audit snippet marks the first place the term appears in the turn, which is not always the hit that was counted (e.g. a turn with several questions). Affected rows were labeled from the real hit and noted.

3. src/episode_dyads.py

3.1 Purpose

Assigns every episode its host→guest gender dyad, using all 418 episodes instead of the balanced 50-per-dyad sample used in v1 (balanced_dataset.py, v1 Section 2, stays available for robustness checks).

3.2 Methodology

- From speaker_features_v2.csv, take the gender of the HOST and of the GUEST speaker with the most words in each episode.
- Apply manual guest-gender corrections for the two episodes where gender detection returned "unknown" (ep_174 and ep_261, both male after review).
- Keep episodes whose dyad is one of MALE->MALE, MALE->FEMALE, FEMALE->MALE, FEMALE->FEMALE (host gender first); none had to be dropped.

3.3 How It Was Run

```
python src/episode_dyads.py --features results/features/speaker_features_v2.csv --out results/features/episode_dyads.csv
```

3.4 Output

Output file: results/features/episode_dyads.csv — 418 rows (episode_id, host_gender, guest_gender, dyad).

Dyad	Episodes
MALE->MALE	226
MALE->FEMALE	67
FEMALE->MALE	64
FEMALE->FEMALE	61
Total	418

3.5 Who the Hosts Are

The corpus has only 9 hosts, identified by matching the host's voice to the speaker library (data/speaker_library.json). Every host has one gender, so all female-host episodes come from 4 hosts and all male-host episodes from 5:

Host	MM	MF	FM	FF	Total
Faye Dsouza	0	0	19	14	33
Karishma Mehta	0	0	28	32	60
Pooja Dhingra	0	0	4	5	9
Soha Ali Khan	0	0	1	1	2
Nikhil Kamath	15	0	0	0	15
Prakhar Gupta	18	5	0	0	23
Raj Shamani	117	32	0	0	149
Ranveer A	49	26	0	0	75
Samdish Bhatia	19	4	0	0	23
Not matched	8	0	12	9	29

MM = MALE->MALE, MF = MALE->FEMALE, FM = FEMALE->MALE, FF = FEMALE->FEMALE. One host, Raj Shamani, hosts 149 episodes, including 117 of the 226 MALE->MALE ones. In 29 episodes the host was labeled by the fallback method rather than a library match, so their identity is not recorded.

3.6 Notes & Limitations

- The dyads are unbalanced (226 MALE->MALE against 61-67 in each other group). The tests in Sections 5-6 are rank-based and do not assume equal group sizes.

- Because host gender is tied to 9 individual hosts and their shows, any difference between male-host and female-host dyads may be about those particular hosts rather than gender. Section 6 adds a test that holds the host fixed.

4. src/dyad_analysis_v2.py

4.1 Purpose

Builds the episode-level dyad dataset from the v2 features (the v2 counterpart of v1 Sections 3 and 4), with descriptive summaries by dyad and by host. The statistical tests on this dataset are separate scripts in src/stat_analysis/ (Sections 5 and 6).

4.2 Methodology

- Join speaker_features_v2.csv to episode_dyads.csv, and put each episode's host and guest values side by side (one row per episode).
- Metrics, all per 1,000 words: hedges, boosters, factuality and deference for both host and guest; direct questions for the host only (Section 2.5). Plus host speaking share (host words / host + guest words) and dominance ratio (host words / guest words).
- No host-minus-guest asymmetry columns. Because booster and deference precision differs between hosts and guests (Section 2.5), subtracting one from the other would mix two different error rates. Each role is instead compared across dyads, e.g. host hedging with male guests vs. female guests.
- Host identity (host_name) is read from the speaker-library match in each transcript.

4.3 How It Was Run

```
python src/dyad_analysis_v2.py
```

All paths default to the files above.

4.4 Output Files

File	Contents
results/dyads/dyad_analysis_v2.csv	418 rows: dyad, host name, host/guest metrics
results/dyads/dyad_median_mad_stats_v2.csv	Median, MAD, n per dyad per metric
results/dyads/host_level_medians_v2.csv	Median of each metric for each of the 9 hosts

4.5 Medians by Dyad

Metric	MM	MF	FM	FF
Host speaking share	0.22	0.28	0.18	0.20
Host hedges	5.04	5.46	5.04	4.64
Guest hedges	4.37	4.86	8.21	7.13
Host boosters	3.05	3.51	5.28	5.98

Guest boosters	2.47	3.46	4.70	6.11
Host factuality	0.96	1.19	1.55	1.28
Guest factuality	0.74	0.96	1.03	1.75
Host deference	0.00	0.24	0.00	0.00
Guest deference	0.00	0.00	0.00	0.00
Host direct questions	3.33	5.15	6.36	3.70

Deference medians are 0 in every dyad because most speakers never use it; the share of episodes where the host uses it at least once is MM 46%, MF 60%, FM 5%, FF 10%.

4.6 Medians by Host

Host	Sex	Eps.	Host share	Host hedges	Host boosters	Host fact.	Host dir. Q	Guest boosters
Faye Dsouza	F	33	0.19	4.12	5.85	1.83	4.15	4.59
Karishma Mehta	F	60	0.18	4.67	5.82	1.09	6.62	5.13
Pooja Dhingra	F	9	0.24	7.53	6.24	2.07	3.43	7.51
Soha Ali Khan	F	2	0.22	8.79	4.24	0.86	2.36	3.42
Nikhil Kamath	M	15	0.22	10.93	3.59	1.03	3.45	6.43
Prakhar Gupta	M	23	0.30	5.31	3.79	1.52	2.38	2.76
Raj Shamani	M	149	0.22	4.14	2.28	0.85	3.88	2.23
Ranveer A	M	75	0.25	7.81	4.04	1.43	3.79	3.17
Samdish Bhatia	M	23	0.22	4.77	3.26	0.47	4.96	2.79

Hosts of the same gender differ as much as the two genders do. For example, host hedging ranges from 4.1 to 10.9 per 1,000 words among male hosts, and from 4.1 to 8.8 among female hosts. This is why the host-gender comparisons in Sections 5-6 need to be read as comparisons between these particular hosts.

5. src/stat_analysis/kruskal_wallis_test_v2.py

5.1 Purpose

Tests whether each metric in results/dyads/dyad_analysis_v2.csv (Section 4) differs across the four gender dyads at all, before asking which dyads differ (Section 6). This is the v2 counterpart of v1 Section 8.

5.2 Methodology

- For each of the 11 metrics in Section 4.2, run scipy's `kruskal()` across the four dyad groups (non-parametric, rank-based, no normality assumption).

- Correct the 11 p-values for multiple testing with the Benjamini-Hochberg false discovery rate (FDR), and call a metric significant if its corrected p-value is below 0.05. (v1 Section 8 did not correct across metrics.)
- Save H, raw p, corrected p, and the median of each dyad to results/stat_analysis/kruskal_wallis_v2_results.csv.

5.3 How It Was Run

python src/stat_analysis/kruskal_wallis_test_v2.py

Paths default to the files above. The Benjamini-Hochberg correction is in src/stat_analysis/fdr.py, shared with the Section 6 scripts (statsmodels is not installed in the convokit environment).

5.4 Results

Metric	H statistic	p-value	FDR p	Significant
Host speaking share	50.50	<0.0001	<0.0001	Yes
Dominance ratio	50.50	<0.0001	<0.0001	Yes
Host hedges	3.26	0.3528	0.3528	No
Guest hedges	50.44	<0.0001	<0.0001	Yes
Host boosters	96.56	<0.0001	<0.0001	Yes
Guest boosters	74.27	<0.0001	<0.0001	Yes
Host factuality	12.68	0.0054	0.0059	Yes
Guest factuality	50.76	<0.0001	<0.0001	Yes
Host deference	65.54	<0.0001	<0.0001	Yes
Guest deference	20.25	0.0002	0.0002	Yes
Host direct questions	34.74	<0.0001	<0.0001	Yes

10 of the 11 metrics differ significantly across dyads. Only host hedging does not ($p = 0.35$): hosts hedge at similar rates in all four dyads.

5.5 Notes & Limitations

- Host speaking share and dominance ratio are two transformations of the same word counts, so their rank-based results are identical; they are one finding, not two.
- A significant result means at least one dyad differs. Because host gender is tied to 9 specific hosts (Section 3.5), a significant host-side metric can reflect differences between those hosts rather than gender. Section 6 separates the two for guest gender.

6. src/stat_analysis/Mann_Whitney_test_v2.py and within_host_test_v2.py

6.1 Purpose

Identifies which pairs of dyads differ for each significant metric (the v2 counterpart of v1 Section 9), and then checks which differences survive when the host is held fixed. Since each host interviews both female and male guests, guest-gender effects can be tested within each host's show; host-gender effects cannot.

6.2 Methodology

- Pairwise (Mann_Whitney_test_v2.py): for each of the 10 metrics significant in Section 5 (read from `kruskal_wallis_v2_results.csv`), run a two-sided Mann-Whitney U test on all 6 dyad pairs, correct the 6 p-values with Benjamini-Hochberg FDR within each metric (as in v1 Section 9), and compute the rank-biserial correlation as effect size. Note the sign: here a positive value means dyad A is higher, the opposite of the convention in v1 Section 9.
- Within host (within_host_test_v2.py): episodes are ranked within each host (as a percentile, 0-1) on each metric. For each host, compute the mean rank of female-guest episodes minus the mean rank of male-guest episodes, then average across hosts, weighting hosts with more of both guest genders more heavily. A value of +0.15 means female-guest episodes sit, on average, 15 percentile points higher within their host's episodes.
- The p-value comes from a permutation test: guest gender is shuffled within each host 10,000 times, and p is the share of shuffles with an effect at least as large. P-values are FDR-corrected across the 10 metrics. Episodes without a library-matched host are left out (8 hosts with both guest genders, 374 episodes).
- Exploratory: the same within-host effect is computed separately for male and female hosts, and their difference is tested with the same shuffles (does the guest-gender effect depend on the host's gender?).

6.3 How It Was Run

```
python src/stat_analysis/Mann_Whitney_test_v2.py
```

```
python src/stat_analysis/within_host_test_v2.py
```

Mann_Whitney_test_v2.py must run after `kruskal_wallis_test_v2.py`, since it tests only the metrics that were significant there. `within_host_test_v2.py` takes `--n_perm` (default 10,000) and `--seed` (default 42) for the permutation test. Outputs: `results/stat_analysis/pairwise_dyad_tests_v2.csv` and `results/stat_analysis/within_host_guest_gender_v2.csv`.

6.4 Results: Pairwise Differences Between Dyads

Of the 54 pairwise tests (9 metrics × 6 pairs, leaving out dominance ratio, which repeats speaking share), 38 are significant after FDR correction. Each cell shows the rank-biserial effect size for dyad A vs. dyad B (positive = A higher); * marks a significant pair.

Metric	MM vs MF	MM vs FM	MM vs FF	MF vs FM	MF vs FF	FM vs FF
Host speaking share	-0.37*	+0.32*	+0.18*	+0.65*	+0.54*	-0.16
Guest hedges	-0.15	-0.48*	-0.40*	-0.42*	-0.32*	+0.07
Host boosters	-0.17*	-0.50*	-0.72*	-0.39*	-0.62*	-0.19
Guest boosters	-0.23*	-0.44*	-0.62*	-0.28*	-0.49*	-0.23*
Host factuality	-0.07	-0.24*	-0.21*	-0.18	-0.14	+0.04
Guest factuality	-0.18*	-0.28*	-0.57*	-0.13	-0.45*	-0.23*
Host deference	-0.14	+0.40*	+0.38*	+0.53*	+0.51*	-0.05
Guest deference	-0.01	+0.25*	+0.22*	+0.26*	+0.24*	-0.03
Host direct questions	-0.27*	-0.44*	-0.12	-0.21	+0.16	+0.33*

The biggest differences separate male-host from female-host dyads (MM and MF vs. FM and FF): boosters are higher in female-host episodes for both hosts and guests, factuality is higher in female-host episodes than in MALE->MALE ones, and host deference is higher in male-host episodes. Host speaking share is highest in MF episodes (median 0.28, vs. 0.18-0.22 elsewhere). Host direct questions are highest in FM episodes.

6.5 Results: Guest Gender Within Host

Metric	Effect (female – male guests)	p (perm.)	FDR p	Significant	Hosts in same direction
Host speaking share	+0.149	<0.0001	0.0003	Yes	6 of 7
Host hedges	+0.013	0.6898	0.8623	No	5 of 7
Guest hedges	+0.055	0.1010	0.1683	No	5 of 7
Host boosters	+0.085	0.0118	0.0236	Yes	6 of 7
Guest boosters	+0.154	<0.0001	0.0003	Yes	7 of 7
Host factuality	-0.001	0.9871	0.9871	No	5 of 7
Guest factuality	+0.131	<0.0001	0.0003	Yes	6 of 7
Host deference	+0.067	0.0111	0.0236	Yes	5 of 7
Guest deference	+0.004	0.9044	0.9871	No	3 of 7
Host direct questions	+0.025	0.4584	0.6548	No	3 of 7

"Hosts in same direction" counts the hosts whose female-guest episodes rank higher, out of the 7 hosts with at least 4 episodes of each guest gender (Soha Ali Khan, with one of each, is left out of the count but not the test).

6.6 Results: Does the Guest-Gender Effect Depend on Host Gender? (Exploratory)

Metric	Effect, male hosts	Effect, female hosts	p (perm.)	FDR p
Host speaking share	+0.169	+0.112	0.4196	0.9332

Host hedges	+0.028	-0.015	0.5464	0.9332
Guest hedges	+0.106	-0.043	0.0359	0.1795
Host boosters	+0.093	+0.070	0.7499	0.9332
Guest boosters	+0.151	+0.157	0.9332	0.9332
Host factuality	+0.002	-0.005	0.9299	0.9332
Guest factuality	+0.120	+0.152	0.6605	0.9332
Host deference	+0.092	+0.021	0.1283	0.4276
Guest deference	-0.003	+0.016	0.7240	0.9332
Host direct questions	+0.145	-0.205	<0.0001	0.0010

Only host direct questions differ after correction: male hosts ask female guests more direct questions than male guests, while female hosts ask female guests fewer. This test was added after the per-host results showed the pattern, so it should be confirmed before being reported as a finding.

6.7 Interpretation: What These Results Suggest

- **Hosts take more of the floor with female guests.** Holds within host: hosts talk more when the guest is a woman (+0.15, in 6 of 7 hosts). The pooled result (MF highest) is therefore not just a difference between shows.
- **Female guests speak with more emphasis and certainty, not more hedging.** Within the same show, female guests use more boosters (+0.15, all 7 hosts) and more factuality markers (+0.13) than male guests. They do not hedge more: the pooled difference in guest hedging (higher in female-host dyads) disappears within host, so it came from the shows, not the guests' gender.
- **Hosts adjust slightly to female guests.** Within host, hosts use more boosters (+0.09) and more acknowledgements (+0.07) with female guests. Host hedging and host factuality do not change with guest gender.
- **Host-gender differences cannot be separated from who the host is.** Female-host episodes have more boosters and factuality, and male-host episodes more deference, but these are differences between 4 and 5 individual hosts, and hosts of the same gender vary as much as the genders do (Section 4.6). They cannot be attributed to host gender with this corpus.
- **Direct questions may follow a cross-gender pattern.** Pooled, FM episodes have the most host direct questions, which agrees with v1 Section 9's finding that female hosts ask more (host_ask), but FF episodes do not. The exploratory within-host result suggests hosts ask more direct questions of guests of the other gender. This needs confirming.

6.8 Notes & Limitations

- The within-host test answers a narrower question than the pooled test: whether the same host behaves differently with female and male guests. It cannot test host gender, because no host appears in both host-gender groups.

- These are associations in observational data. Guests are not randomly assigned to hosts, so female and male guests on the same show may also differ in profession or topic (compare the topic-control analyses), which could explain part of the guest-gender effects.
- Feature precision is 0.71-0.86 (Section 2). Errors that are the same across dyads weaken differences rather than create them; errors that differ by guest gender could not be ruled out with the small number of female speakers in the audits.
- FDR correction is applied within each test family (11 omnibus metrics; 6 pairs per metric; 10 within-host metrics), not across all tests together.